

Distributed Databases

Exercise 01:

Let us consider a database made up of three tables: Customer, Product and Sales. The Sales table contains the foreign keys of the other two tables as well as to the quantity sold and the unit price of the sale (UP).

Sales			
CID	PID	Qty	UP
616	106	5	25
616	104	10	50
515	104	5	10
414	105	2	14
212	106	4	14
111	101	5	20
212	101	9	100
313	102	3	200
414	102	22	102
515	102	2	100
515	103	4	17
212	103	6	45
111	105	89	44
212	104	8	40
515	104	7	20
616	104	5	20
212	105	2	10
212	105	2	10
212	106	4	18
313	105	4	19
313	106	6	15

Customer		
CID	NameC	City
616	Gilles	Alger
515	Yves	Blida
414	Yamine	Tipaza
313	Ladjel	Tipaza
212	Eric	Alger
111	Pascal	Alger

Product		
PID	NameP	Type
106	Sonoflore	Beauty
105	Clarins	Beauty
104	WebCam	Multimédia
103	Barbie	Toys
102	Engrais	Gardening
101	SlimForm	Fitness

Let us consider the following queries:

- Q1: Select NameC from Customer where City = “Algiers”
- Q2: Select * from Customer where City =”Blida”
- Q3: Select NameP from Product where Type=”Beauty”

1. Give the execution result of each query.
2. What is the best horizontal partitioning of the Customer and Product tables so that each query loads only the necessary tuples.
3. Give the fragments obtained for each table.
4. Propose a rewriting of the three queries according the obtained fragments.

Let's assume that the distribution of customer cities is as follows: Algiers: 800,000, Tipaza: 190,000 and Blida 10,000.

By considering that the cost of executing a query corresponds to the number of pages to load from secondary memory (we assume that a page can store 10 tuples)

1. What is the cost of executing Q1 and Q2 before portioning.
2. What is the cost of executing Q1 and Q2 after portioning.

3. What conclusion can you draw?.

Now, let us assume the following query:

• Q4: Select count(*) from Customer c, Sales s where s.CID=c.CID and c.City="Blida"

1. What is the best way to fragment the customer and sales tables so that Q4 only loads the necessary tuples.
2. Give the obtained fragments.
3. Propose another rewrite for the query Q4.

Let us assume another query Q5:

• Q5 : Select count(*) from Customer c, Product P, Sales s where s.CID=c.CID and s.PID=p.PID and c.Ville = "Blida" and s.type= "Beauty"

1. What is the best way to fragment the customer and sales tables so that Q5 only loads the necessary tuples.
2. Give the obtained fragments.
3. Propose another rewrite for the query Q5.
4. What does the number of fragments of the sales table represent compared to the number of Product and Customer fragments?

Exercise 02

A driving school, established in Algiers for several years, has not stopped developing on the national territory. Therefore, its leaders decided to create agencies in the cities of Constantine and Oran.

The centralized database that was initially developed has the following structure:

AGENCY (NumAG, cityAG, Num-manager*, Nbr-instructors)

ADMINISTRATIVE_STAFF (NumS, AddressS, NameS, NumAG*, Date of birth, Function)

VEHICLE (NumV, Brand, Type, NumAG*, Num-Instructor*)

CUSTOMER (NumC, NameC, AddressC, NumAG*, Age)

COURSE (Num-Instructor*, NumC*, Course-Date, NumV*)

EXAM (NumC*, ExamDate, Result)

Remarks :

- The manager and instructors are part of the administrative staff.
- A vehicle is assigned to an instructor
- The school wants to create three sites to host data relating to the agencies of each city: Algiers, Oran and Constantine Sites.

1. Fragmentation

- a) Propose a fragmentation scheme for this database taking into account hypotheses related to sites. Give the algebraic expressions for each table.
- b) What are the fragmentation possibilities of the COURSE table. Give the best solution.
- c) Give the SQL queries corresponding to the fragmentation of the relations AGENCY and Customer.
- d) Oracle fragments a table when it is created. Give a solution to fragment a table already containing data. Apply the solution to the AGENCE table

2. Allocation

- a) Propose an allocation scheme for all fragments.
- b) Vehicles belonging to agencies in the wilaya of Algiers are very often accessed from Oran.

- Propose an improvement to the allocation scheme by taking this constraint into account.
- Give the disadvantages of this solution.

3. Evaluation of distributed queries

Knowing that the fragmentation is transparent to the user and that the following query (Q), is issued from the Algiers site: "Give the name and number of all customers under the age of 25, belonging to the agencies of Algiers or 'Oran, and have a positive result in the exam on 05/22/2019'

- a. Give the associated SQL script, the global algebraic tree, the canonical tree as well as the simplified algebraic tree.
- b. Give two strategies for executing the query Q.

Additional Exercise:

Let us consider a database related to a mobile telephone operator, composed of two tables: Subscriber and Call, where the logical schema is as follows.

Subscriber (NumTel, SurnameS, FirstNameS, Age, Region, Professional Situation, Address).

Call (CallerNumTel, Date, Time, CorrespondentNumTel*, Duration, Amount).*

The management of this operator's offers is based on two criteria: the region where the customer lives and their professional situation. A Customer's region can be: North, South, East or West. Their professional situation can be: Student, Employee, retired, etc.

The dealership has three departments located as follows: after-sales (Site A), marketing (Site B) and technical (Site C). The marketing department wants to create an offer for each region of the country, therefore it wants to group customers from each region in the same partition.

1. How can we fragment the Subscriber table to satisfy this need. Give the algebraic expressions.
2. Give the SQL query carrying out this fragmentation.
3. An after-sales service employee wants to find the average amount paid by customers from the south and the north on 01/01/2021.
 - a. Rewrite the query after fragmentation.
4. The marketing department wants to fragment the table *Call* in the same way as the *Subscriber* table.
 - a. What is the type of this fragmentation?
 - b. Give the algebraic expressions used to define those fragments
 - c. Give the SQL query that fragments the table *Call*.
5. The operator has four regional offices (directorates): Oran, Algiers, Ouargla and Constantine. It distributes the data on each directorate in such a way as to have in each directorate the Subscribers living in the same region as well as their calls.
 - a. Give the corresponding allocation scheme: the fragments in each directorate.
 - b. A query Q is launched in Algiers. It returns the average consumption of customers living in the south or west region whose age is less than 25 years old.
 - i. Give the corresponding request
 - ii. Identify all execution strategies for this query.